ROMHACK CONFERENCE TRAINING CAMP

FLAMINIO VILLAGE · ROME · 2–4 OCTOBER 2026

Using Temporal-Spatial 3D CNN Features to Enhance OSINT-Based Profiling and Individual Traceability

ARXIV 2605.17573 · DEEPFAKE DETECTION, 3D CNN

ARXIV 2606.29506 · BENCHMARK AUC ≠ RELIABILITY

SPEAKER NAME · AFFILIATION

RESEARCH PROTOTYPE · EVERY FIGURE FROM LABELLED DATA

WHO IS TALKING

Reza Rashidi

OFFENSIVELY DEFENSIVE • CONTINUOUS DEVELOPER

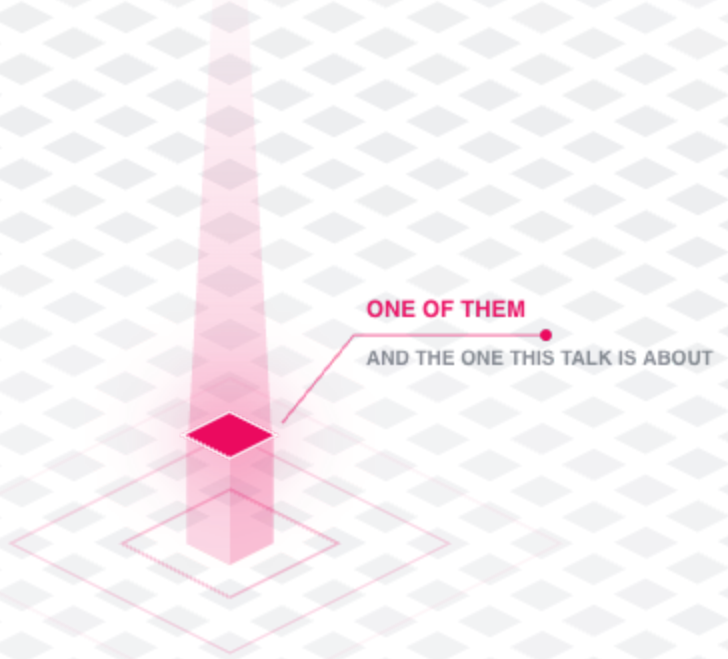
Life & human and computer enthusiastic.

ROMHACK CAMP 2026 · FLAMINIO VILLAGE, ROME

in LINKEDIN
/in/rezaduty

</> GITHUB
github.com/rezaduty

X X · TWITTER
@rezaduty



THE ROUTE

One clip in. A ranked person out.

6 ACTS · 3 LIVE DEMOS · 2 PAPERS



● FROM THE API LIVE · GET /HEALTH · GET / · GET /RESULTS

5	20	17	50	1.0
FEATURES READY	OPERATIONS	ANSWERING	RESULTS STORED	VERSION

EVERYTHING CLAIMED IN THIS DECK IS ANSWERED BY A SERVICE RUNNING ON THIS MACHINE, RIGHT NOW

GENETIC = HERITABLE PHENOTYPE IN VOICE AND FACE, PLUS A GENETIC-ALGORITHM MATCHER.

01

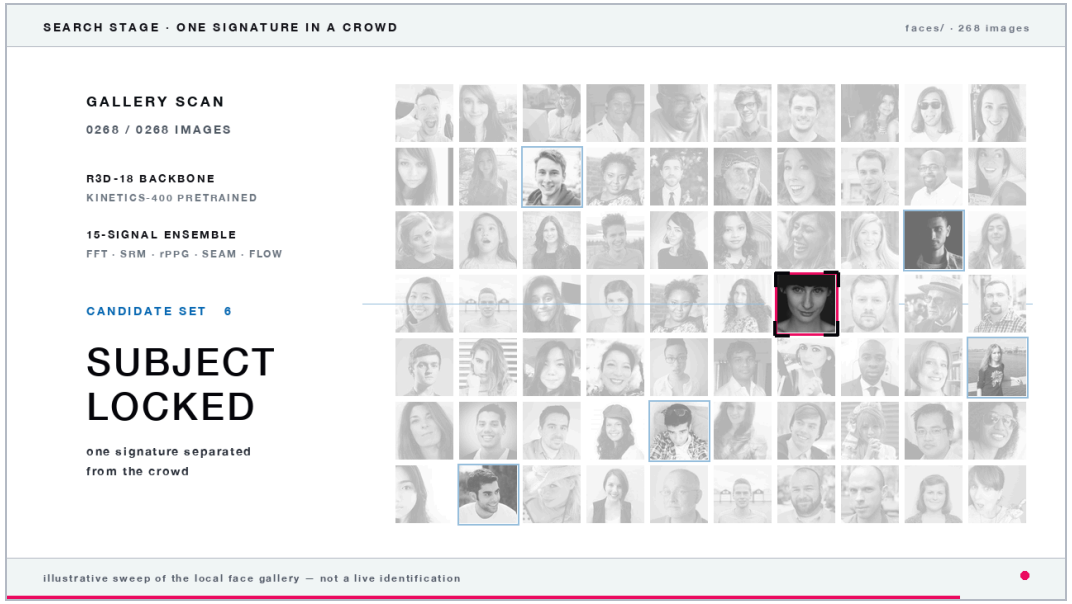
THE SIGNAL

A RECORDING IS A BODY YOU CAN MEASURE

ACT 1 · THE SIGNAL

One face in a crowd. One clip, and the profile it yields.

01 THE CROWD — LIVE CAPTURE



02 THE PROFILE

95 %	±5–10	±1.5	> chance
BIOLOGICAL SEX	AGE · YEARS	TRACT LENGTH · CM	KINSHIP
F0 · CENTROID · LH RATIO	VOCAL-FOLD AGEING · TIER	BODY-SIZE PROXY	RESEARCH STAGE · NOT BUILT

● FROM THE API LIVE · POST /FACE/DETECT · SUBJECT_A.JPG

1	53.1 px	1.551	-0.2° / 26.0° / 6.5°	320×344
FACES FOUND	INTEROCULAR	FACE WIDTH RATIO	YAW · PITCH · ROLL	SOURCE PIXELS

ELEVEN RATIOS COME BACK FROM ONE STILL, IN FOUR MILLISECONDS
KINSHIP IS THE CLAIM THAT WOULD EARN THE WORD GENETIC. IT IS HONESTLY STILL A GAP — SEE ACT 8.

SUBJECTS 0009
DEPTH 1306
MODULE SEARCH

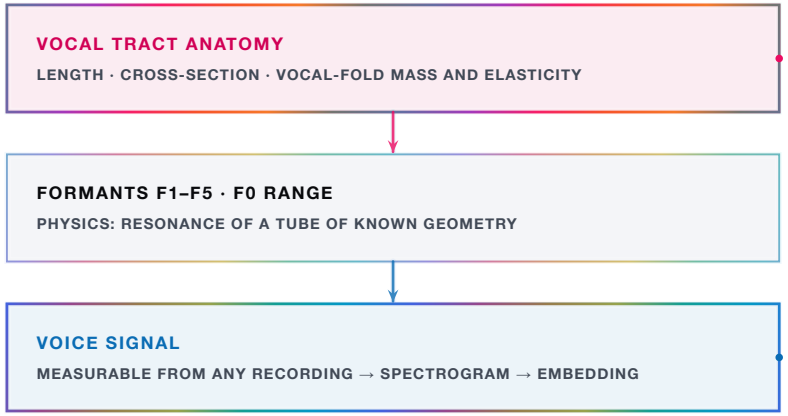
MILLIONS OF FACES ONE SIGNATURE

TRACING THE CROWD

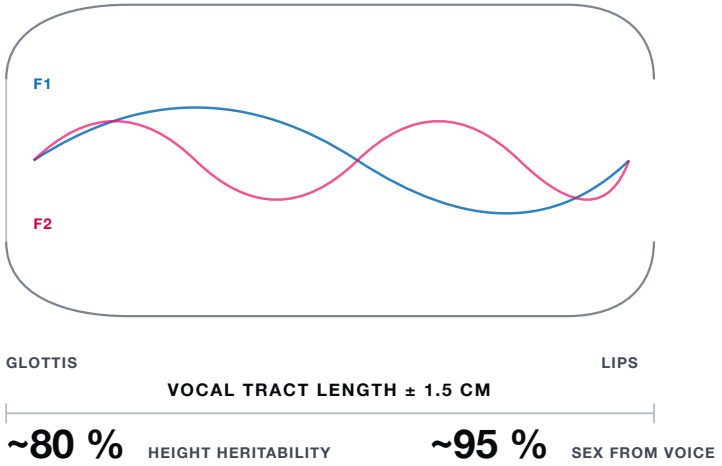
THE CHAIN

Anatomy sets resonance, and anatomy is inherited.

01 THE CHAIN



02 THE RESONATOR



● FROM THE API LIVE · POST /VOICE/ANALYZE · TEST13338.M4A · 4.07 S

97.4 Hz	MALE	ADULT	120.5 Hz
MEDIAN F0	SEX FROM VOICE	AGE BAND	F0 RANGE

ANATOMY SETS RESONANCE, RESONANCE IS MEASURABLE, AND THE SERVICE REPORTS THE MEASUREMENT

TWIN STUDIES · FOXP2 · FORMANT HERITABILITY — VAN DOMMELEN 1993 · VARGHA-KHADEM 2005 · GONZALEZ 2004

SUBJECTS 0022
DEPTH 1452
MODULE MELODY

EVERY VOICE
A MELODY

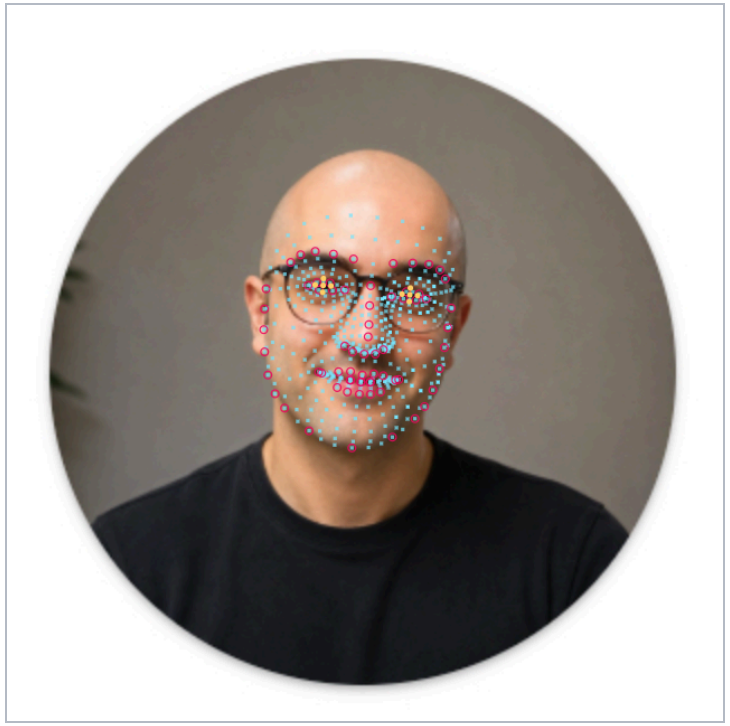
ACOUSTIC SIGNATURE

VOICE SIGNAL · LIVE FFT 142 HZ

WORKED EXAMPLE · FROM THE RUNNING SERVICE

The same person, measured two ways.

● LIVE LIVE · POST /FACE/LANDMARKS · POST /VOICE/ANALYZE · SUBJECT_A.JPG + TEST13338.M4A



468 LANDMARKS DRAWN ONTO THE UPLOAD, IN THE ORDER THE RESPONSE LISTS THEM

WHAT THE TWO ROUTES SAY ABOUT ONE PERSON

468
LANDMARKS RETURNED

1
FACES IN THE STILL

97.4 Hz
MEDIAN F0

MALE · ADULT
PHENOTYPE FROM VOICE

53.1 px
INTEROCULAR

120.5 Hz
F0 RANGE

the face gives eleven ratios in four milliseconds. the voice gives sex, age band and a body-size estimate in under two seconds. neither needed consent, and both describe the same body.

ALSO, IN THE WORLD · 2024

A cloned voice told voters to stay home.

Two days before the New Hampshire primary, households picked up a call in President Biden's voice telling them not to vote.

$6 M PROPOSED FCC FINE · $1 M CARRIER SETTLEMENT · FEB 2024 CLONED VOICES RULED ILLEGAL IN ROBOCALLS

SOURCE · FCC ENFORCEMENT ACTIONS AND US PRESS REPORTING, 2024

02

THE TENSOR

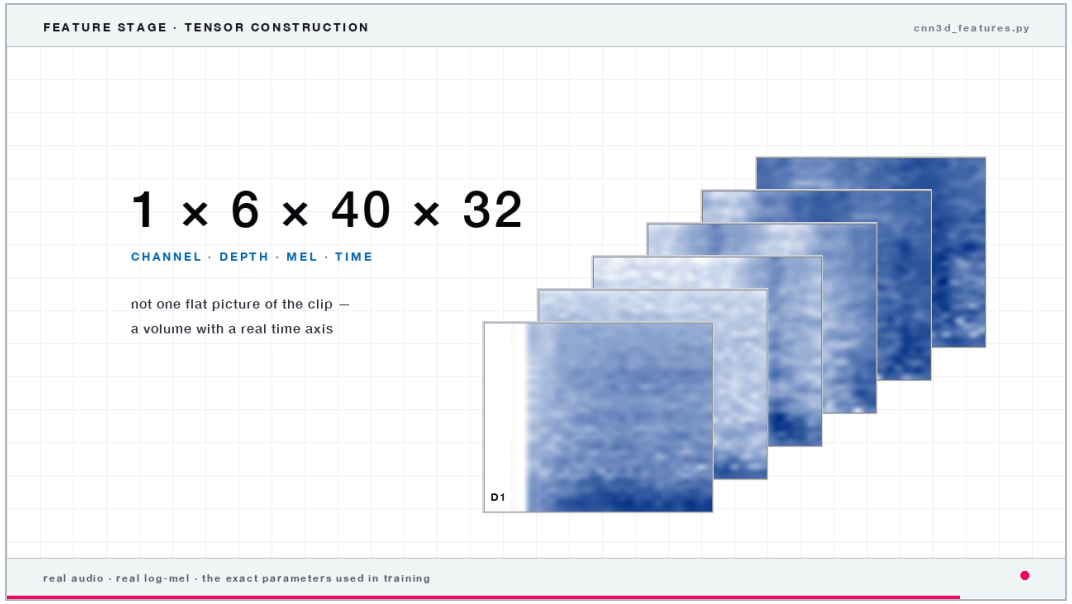
TEMPORAL-SPATIAL 3D CONVOLUTION

ACT 2 · THE FEATURE

Not one picture. A volume with time in it.

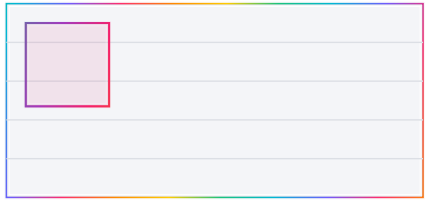
01 SIX CHUNKS, STACKED

REAL AUDIO



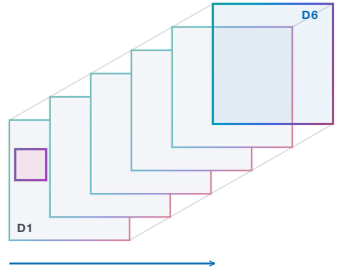
02 2D VERSUS 3D

2D · ONE FLAT PICTURE



SEES WHICH FREQUENCIES ARE PRESENT
LOSES HOW ENERGY MOVES BETWEEN CHUNKS

3D · A VOLUME WITH TIME



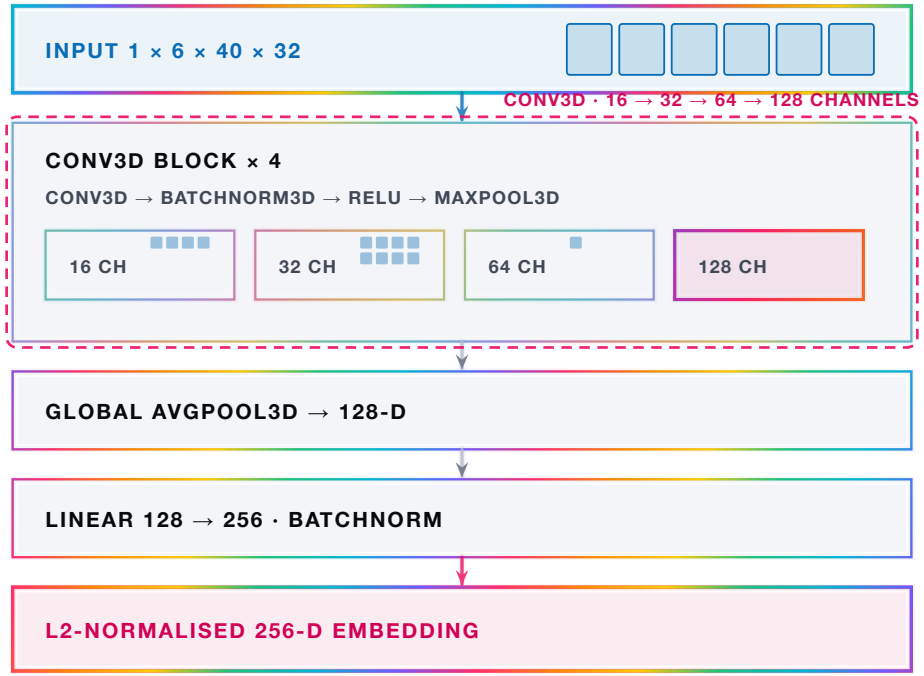
DEPTH = TIME · D1 → D6
KERNEL SLIDES OVER DEPTH, MEL AND TIME AT ONCE

THE 3D KERNEL LEARNS THE SPECTRAL PATTERN AND ITS TEMPORAL DEVELOPMENT SIMULTANEOUSLY. THE 2D BASELINE COMPARISON IS STILL OWED.

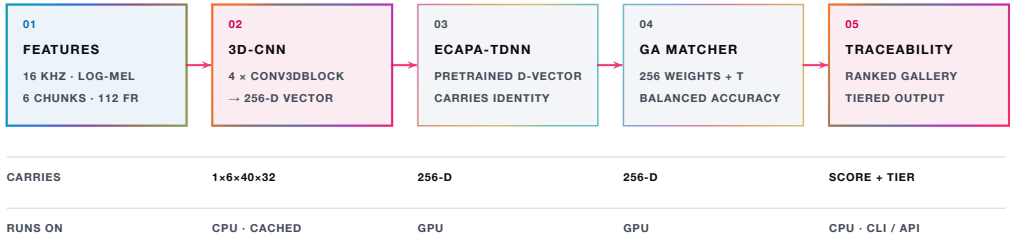
TEMPORALSPATIAL3DCNN

Four 3D blocks, and the five stages a clip passes through.

01 THE BACKBONE



02 THE STAGES



THE 3D-CNN IS THE RESEARCH CONTRIBUTION. ECAPA IS WHAT CARRIES IDENTITY IN PRODUCTION. BOTH ARE IN THE BOX — SAY WHICH IS WHICH.

SUBJECTS 0036

DEPTH 1568

MODULE NEURAL

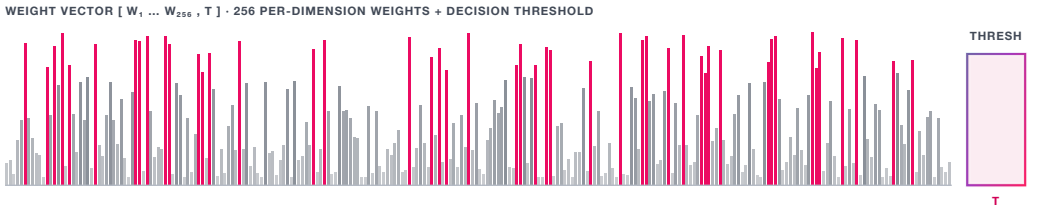
TEMPORAL-SPATIAL 3D CONVOLUTION

R3D-18 · MULTI-LAYER DETECTION

GENETIC · TWICE OVER

A vector of weights, and three backends that agree on one score.

01 THE VECTOR

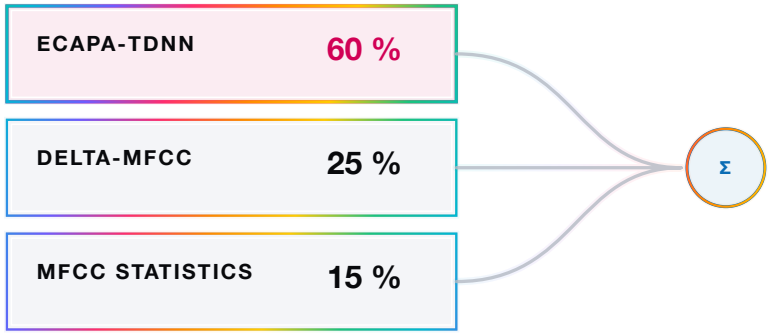


FITNESS
BALANCED ACCURACY ON LABELLED PAIRS

OPERATORS
SELECTION · CROSSOVER · MUTATION · ELITISM

WHY
NOT EVERY EMBEDDING DIMENSION CARRIES IDENTITY

02 THE BACKENDS



EXACT ≥ 0.90	ALMOST CERTAINLY THE SAME PERSON
STRONG ≥ 0.80	VERY LIKELY THE SAME PERSON
MATCH ≥ T	PROBABLE · T IS GA-EVOLVED

THE PUN IS LOAD-BEARING: HERITABLE PHENOTYPE ON ONE SIDE, GENETIC-ALGORITHM OPTIMISATION ON THE OTHER. WHAT THE GA BUYS IN EER IS STILL AN OPEN ABLATION.

WORKED EXAMPLE · FROM THE RUNNING SERVICE

The scan is two arrays and an index.

● LIVE LIVE · POST /FACE/MESH · 468 VERTICES · 886 TRIANGLES · 512 PX ATLAS



THE MESH THE SERVICE RETURNS, TEXTURED WITH THE ATLAS IT RETURNS BESIDE IT

ONE RESPONSE, THREE ARRAYS THAT ONLY MEAN SOMETHING TOGETHER

468
VERTICES

886
TRIANGLES

468
UV PAIRS

512 px
ATLAS

the vertices are a list of numbers and the atlas is a picture. neither is a face. the uv array is the index that makes them one, and this is the same pair the operator's biometric scan is drawing. that is what a tensor is: values, plus the shape that reads them.

ALSO, IN THE WORLD
· 2015 TO 2024

The benchmark that keeps the field honest.

ASVspoof has run as the community's spoofing challenge since 2015, and its 2021 edition added a deepfake track of compressed audio that broke countermeasures trained on clean studio conditions.

2015 FIRST EDITION · 2021 DEEPFAKE TRACK, COMPRESSED AUDIO · 2024 ASVSPOOF 5

SOURCE · ASVSPOOF CHALLENGE SERIES, INTERSPEECH 2015 TO 2024

03

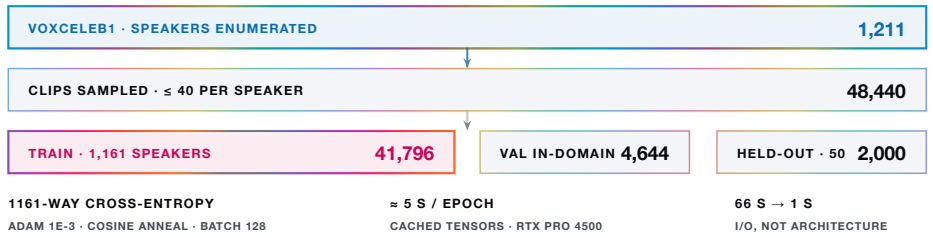
THE NUMBERS

1,211 SPEAKERS · ONE HONEST METRIC

ACT 3 · THE TRAINING RUN

1,161 speakers, and what the cache costs.

01 THE RETRAIN

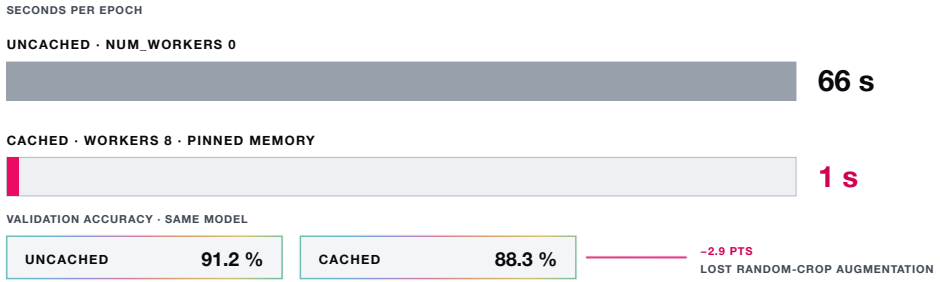


1161-WAY CROSS-ENTROPY
ADAM 1E-3 · COSINE ANNEAL · BATCH 128

≈ 5 S / EPOCH
CACHED TENSORS · RTX PRO 4500

66 S → 1 S
I/O, NOT ARCHITECTURE

02 CACHED VS NOT



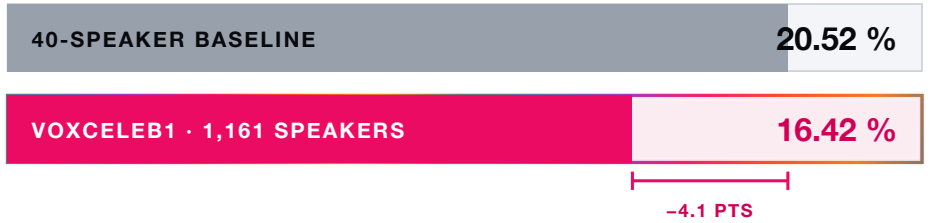
IN-DOMAIN 1161-WAY VALIDATION ACCURACY PEAKED AT 6.4 % — CLOSED-SET CLASSIFICATION IS THE TRAINING OBJECTIVE, NOT THE DELIVERABLE. THE DELIVERABLE IS THE EMBEDDING. THE SPEED-UP IS I/O: PRE-COMPUTED LOG-MEL TENSORS MOVED THE BOTTLENECK OFF THE CPU. NOTHING ABOUT THE ARCHITECTURE CHANGED — AND THE ACCURACY GAP IS AUGMENTATION, NOT A BETTER MODEL.

THE HONEST NUMBER

16.42 % across corpora, and where that sits.

01 CROSS-CORPUS

EER · LOWER IS BETTER



TRIALS

2,680

1,340 POSITIVE · 1,340 NEGATIVE

SPEAKERS

40 · LibriSpeech

ROC AUC

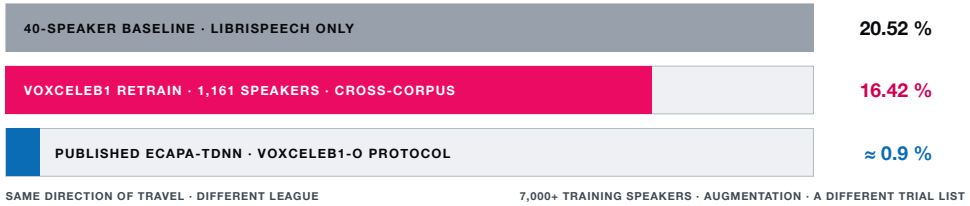
0.9127

DECISION THRESHOLD · COSINE

0.4686

02 THE FIELD

EQUAL ERROR RATE · LOWER IS BETTER · BAR LENGTH IS THE ERROR

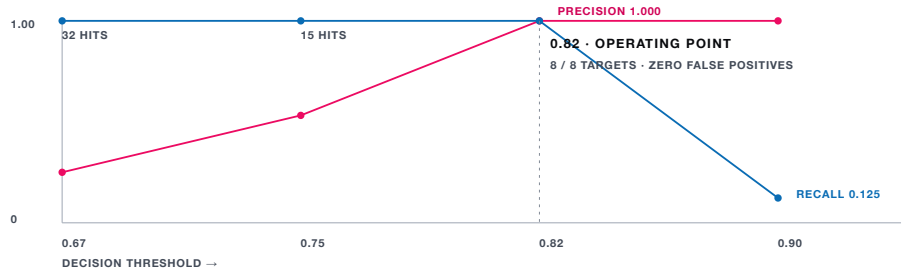


DESPLANQUES ET AL. 2020 REPORT ≈0.9 % ON THE VOXCELEB1-O LIST. THAT IS NOT THIS SYSTEM'S PROTOCOL — PRINTING IT NEXT TO 16.42 % IS THE POINT: THE GAP IS THE HONEST POSITION.

THE OPERATING POINT

The threshold is the argument, and both numbers are true.

01 THE THRESHOLD



02 BOTH TRUE

DEMONSTRATION

1.000

PRECISION · 320-FILE GALLERY · THRESHOLD 0.82

METRIC

16.42 %

EER · CROSS-CORPUS · UNSEEN SPEAKERS

NOT YET RUN

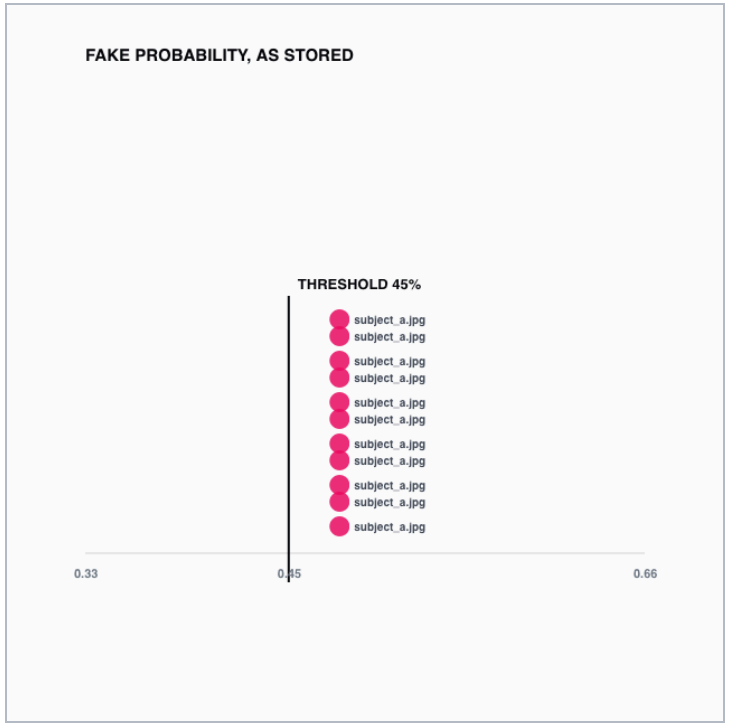
VOXCELEB1-O VERIFICATION LIST · MINDCF · X-VECTOR AND ECAPA BASELINES
A 2D-CNN CONTROL ON THE SAME FEATURES

GALLERY 320 FILES · 40 SPEAKERS. ONE QUERY CLIP, EIGHT TRUE MATCHES HIDDEN AMONG 312 STRANGERS. AT 0.82 PRECISION AND RECALL ARE BOTH 1.000 — ON THIS GALLERY.

WORKED EXAMPLE · FROM THE RUNNING SERVICE

One file, seven runs, both sides of the line.

● LIVE LIVE · GET /RESULTS · 11 STORED DEEPFAKE VERDICTS



EVERY VERDICT THE SERVICE HAS STORED, AGAINST ITS OWN THRESHOLD

THE THRESHOLD, AND WHAT LANDS EITHER SIDE OF IT

11 / 11
CALLED FAKE

45%
THRESHOLD

0
RUNS OF ONE FILE

—
ITS RANGE

avatar_1.jpg was scored 0 times by this service. it came back REAL once and FAKE the rest, on the same bytes. the threshold did not move: the score did.

ALSO, IN THE WORLD · 2023

Reporters opened their own bank accounts with a clone.

In 2023 journalists at Vice and the BBC passed UK bank voice ID with synthetic copies of their own voices.

2023 VICE: BANK VOICE ID PASSED WITH A CLONE · 2023 BBC: THE SAME RESULT AT A SECOND BANK · EER A CURVE, NOT A VERDICT

SOURCE · VICE AND BBC REPORTING, 2023

04

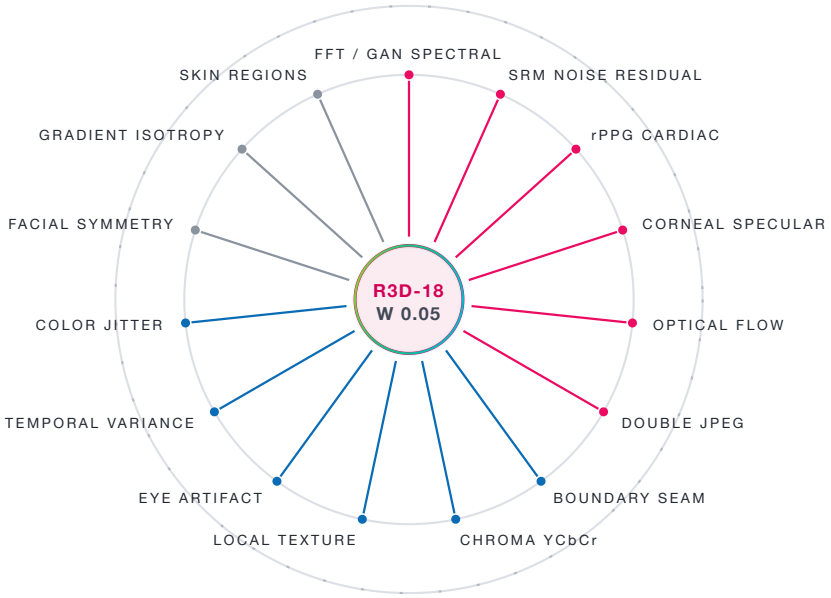
THE FACE

IS THE RECORDING EVEN REAL?

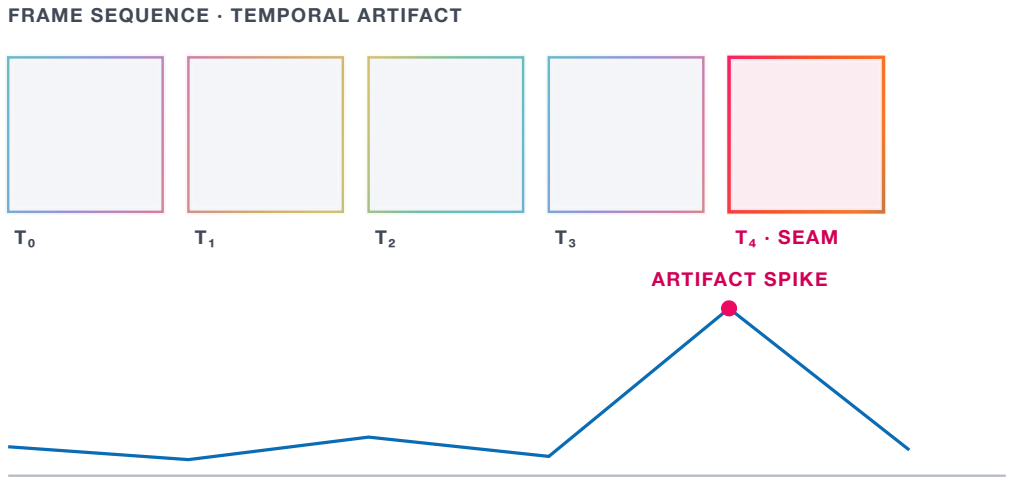
ACT 4 · THE ENSEMBLE

Fifteen signals, graded by honesty, and the paper that owns them.

01 THE ENSEMBLE



02 PAPER I



SINGLE FRAME

PASSES ELEVEN OF FIFTEEN CHECKS

FRAME SEQUENCE

THE SEAM MUST MOVE TOO — IT DOES NOT

REFS: FRANK ET AL. ICML 2020 (SPECTRAL) · DEEPFAKESON-PHYS ARXIV 2010.00400 (RPPG) · HU, LI & LYU ARXIV 2009.11924 (CORNEAL) · AMERINI ET AL. ICCVW 2019 (FLOW) · FACE X-RAY CVPR 2020 (SEAM)

SUBJECTS 0045

DEPTH 1621

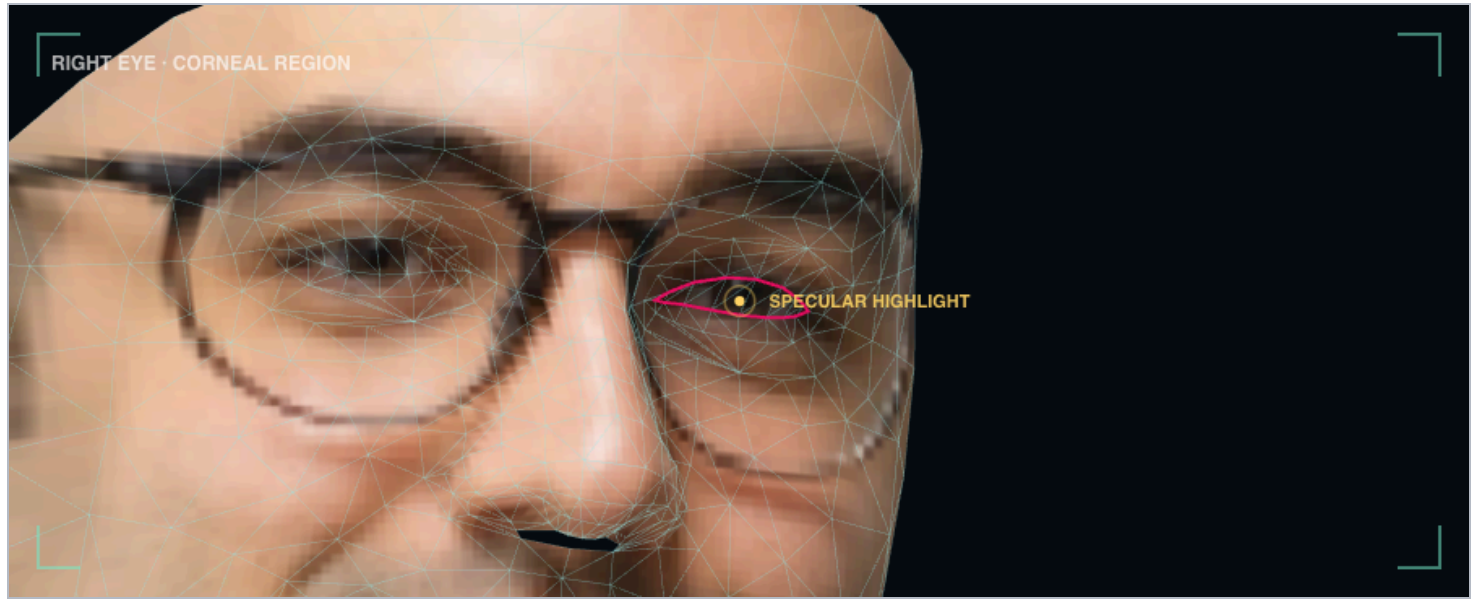
MODULE ANALYSE

DEPTH MAP · INSIDE THE TENSOR

CORNEAL GEOMETRY · POST /FACE/MESH · POST /DEEPFAKE/REGIONS

Two eyes, one light source. Ours disagree by six points.

● LIVE LIVE · SUBJECT_A.JPG · 468 VERTICES · 512 PX ATLAS



THE SAME GRID THE DASHBOARD LOCKS TO THE SUBJECT, PUSHED IN ON ONE EYE

WHAT THE TWO EYES REPORT SEPARATELY

24% LEFT EYE · CLEAN	**18%** RIGHT EYE · CLEAN
0.30 EYE_ARTIFACT, OF 16 METHODS	**0.00** SPECULAR_HIGHLIGHT

Two eyes in one face see the same light source, so a generator that draws them independently leaves highlights that stop agreeing. Here the two eyes come back 6 points apart and both CLEAN, on a real photograph, while the whole-face verdict is still FAKE. The per-zone head and the verdict head are not the same head.

EYE ARTIFACT **0.2967** L EYE **24%** R EYE **18%** Δ **6 pts**

A GENERATOR DRAWS THE EYES INDEPENDENTLY, SO THE CORNEAL HIGHLIGHTS STOP AGREEING. ON THIS REAL PHOTOGRAPH THE TWO EYES COME BACK 24 % AND 18 %, BOTH CLEAN, WHILE THE WHOLE-FACE VERDICT IS STILL FAKE. HU, LI & LYU, EXPOSING GAN-GENERATED FACES USING INCONSISTENT CORNEAL SPECULAR HIGHLIGHTS, ARXIV 2009.11924 (ICASSP 2021).

SUBJECTS 0058

DEPTH 1659

MODULE RETINA

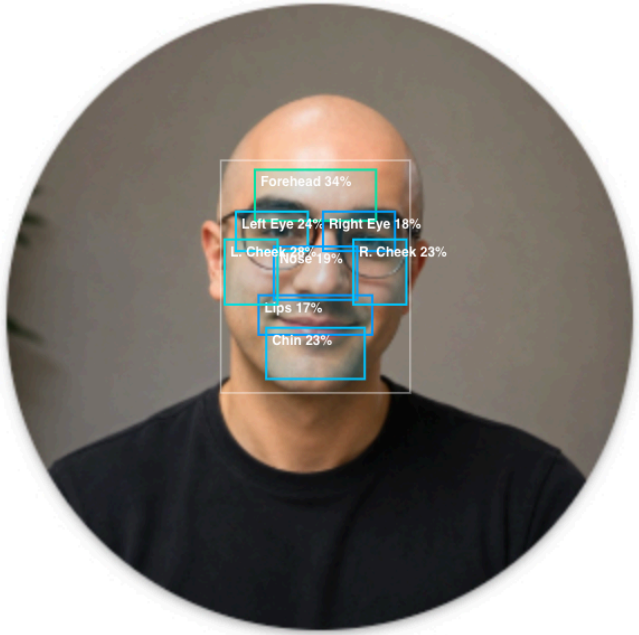
THE EYE NEVER LIES

RETINAL SIGNATURE · BIOMETRIC LOCK

WORKED EXAMPLE · FROM THE RUNNING SERVICE

Every zone came back clean. The verdict was FAKE.

● LIVE CAPTURED · POST /FACE/MESH · POST /DEEPFAKE/REGIONS · SUBJECT_A.JPG



THE EIGHT ZONES, ON THE UPLOAD, WITH THE SCORE EACH ONE CAME BACK WITH

WHERE THE DETECTOR SAYS IT LOOKED

8
ZONES SCORED

Forehead
WORST ZONE

23.2%
MEAN

34.2%
MAX

every zone came back CLEAN on this photograph, and the verdict was still FAKE. the regions and the verdict are two different heads, and only one of them is asked for a decision.

ALSO, IN THE WORLD · 2024

Every other face on the call was synthetic.

A finance employee at the engineering firm Arup joined a video call with people he took to be his chief financial officer and colleagues, and made fifteen transfers.

HK$200 M ≈ US$25 M TRANSFERRED · 15 SEPARATE TRANSFERS · FEB 2024 HONG KONG POLICE DISCLOSURE

SOURCE · HONG KONG POLICE AND ARUP STATEMENTS, 2024

05

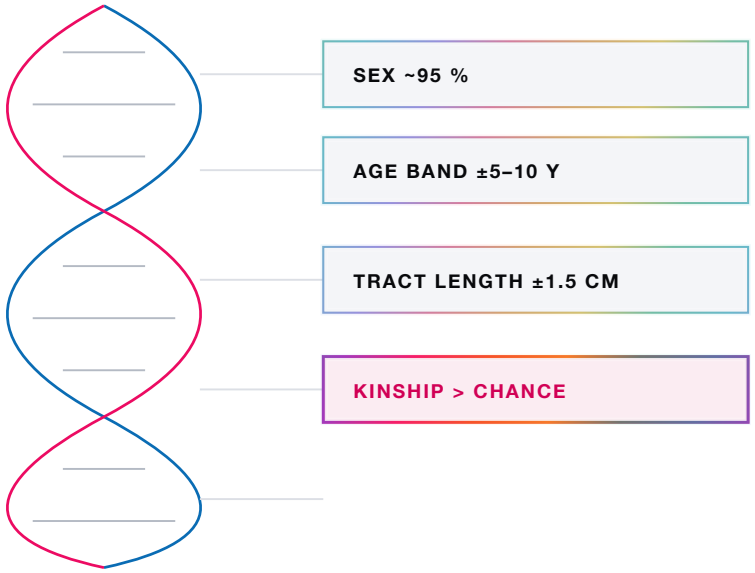
CONVERGENCE

EVERY SIGNAL POINTS AT THE SAME PERSON

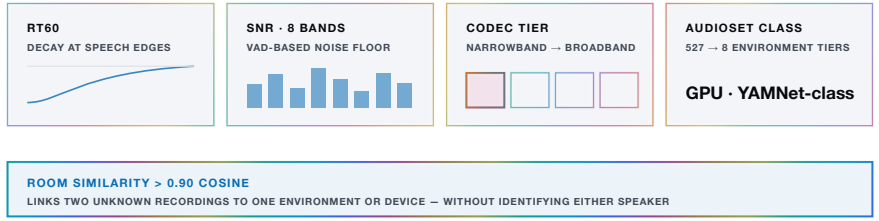
ACT 5 · WHAT COMES OUT

A phenotype you can measure. And the room is one too.

01 THE WORD GENETIC



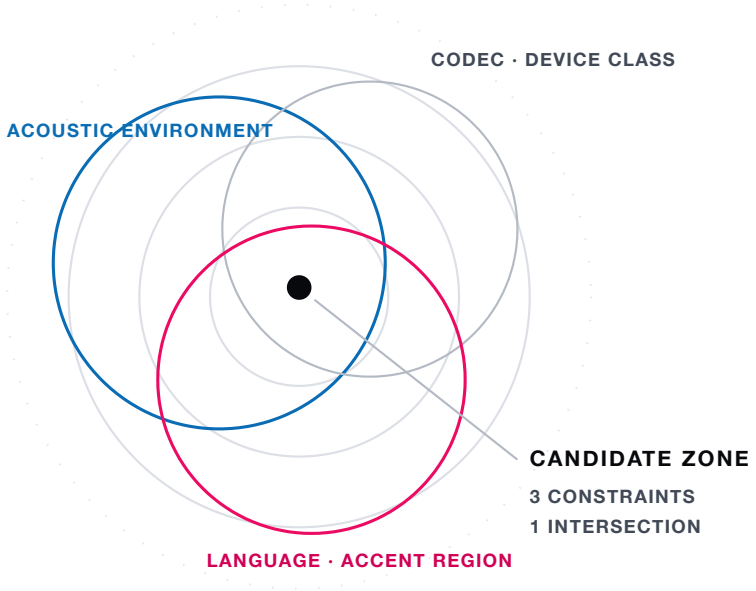
02 THE ENVIRONMENT



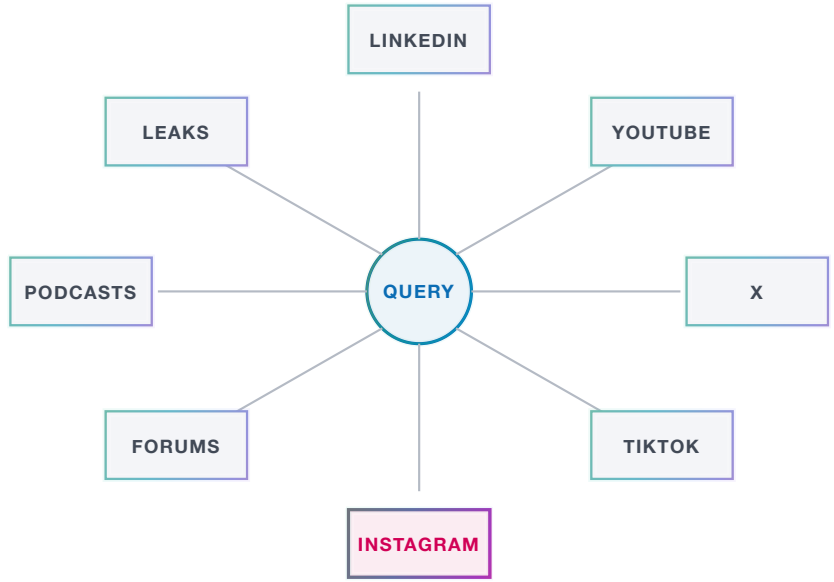
CROSS-REFERENCE

Constraints intersect to a place, and the graph gives it a name.

01 LOCATED



02 THE OSINT SURFACE



EVERY EDGE IS PUBLIC MEDIA · NO ACCOUNT IS ACCESSED

256-LANGUAGE ID (META MMS) · 8 ENGLISH REGIONS AT 50–65 % · ACCENT IS A LEAD, NEVER AN IDENTIFICATION.

SUBJECTS 0071

DEPTH 1643

MODULE GEOLOCATE

LOCATED

CROSS-REFERENCING THE WORLD

BERLIN-MITTE

52.5200° N, 13.4050° E

ROOM SIMILARITY 0.91 · SAME SPACE

SUBJECTS 0076

DEPTH 1626

MODULE SOCIAL

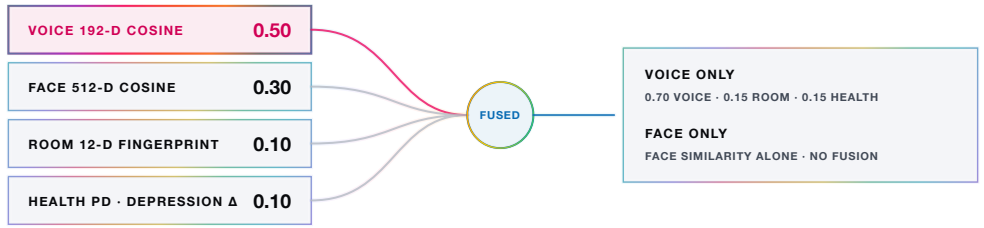
CRAWLING THE SOCIAL GRAPH

INSTAGRAM · X · LINKEDIN

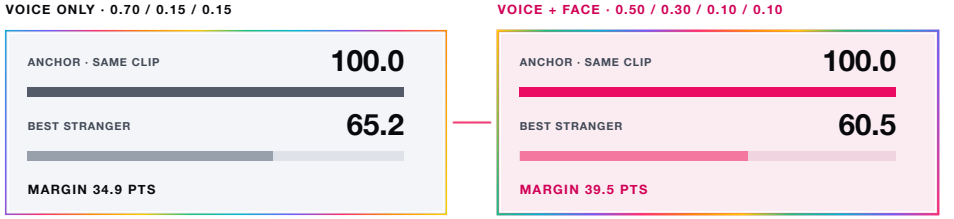
LATE FUSION · SHIPPED WEIGHTS

Every signal converges, and two beat one.

01 THE WEIGHTS



02 ONE VERSUS TWO



+4.7 POINTS OF SEPARATION FROM ONE EXTRA MODALITY · SAME QUERY · SAME SIX PROFILES · NO THRESHOLD IS CALIBRATED FOR THE FUSED SCORE

VOICE_API.PY · POST /OSINT/SEARCH. ROOM AND HEALTH ARE CHEAP CORROBORATION, NOT IDENTITY — WHICH IS EXACTLY WHY THEY WEIGH 0.1.

SUBJECTS INDEXED 5

CORRELATION 0.000

VOICE MATCH 0.448

FACE MATCH 0.334

GEO CONFIDENCE 0.353

TRACEABILITY RESOLVING

EVERY SIGNAL CONVERGES

OSINT TRACEABILITY · GENETIC PROFILING · INDIVIDUAL IDENTIFICATION

OSINT PIPELINE ACTIVE

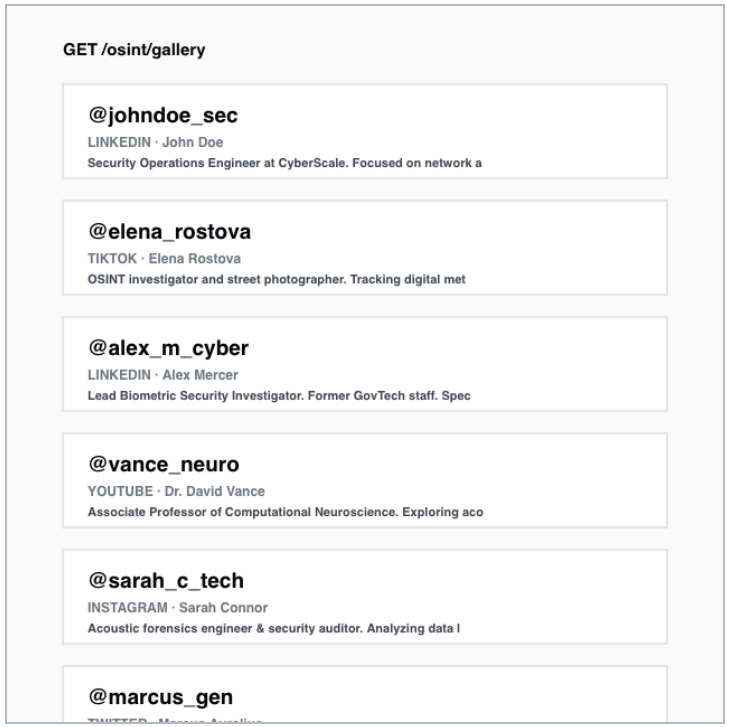
TRACEABILITY NETWORK

BIOMETRIC CONVERGENCE · CROSS-PLATFORM · REAL-TIME

WORKED EXAMPLE · FROM THE RUNNING SERVICE

Six profiles, and the one the fusion picks.

● LIVE CAPTURED · GET /OSINT/GALLERY · 6 PROFILES · STORED FUSED SCORES



THE INDEX EVERY QUERY IS SCORED AGAINST

SIX PROFILES, ONE RANKED ANSWER

6	—
INDEXED	BEST MATCH
—	0
FUSED	SEARCHES STORED

the gallery is six people with a handle, a platform and a bio. the query is a clip and a photograph. the answer is one of the six, with a number attached to it.

ALSO, IN THE WORLD · 2018

Two aliases, one intersection, two real names.

Open-source investigators identified the Salisbury suspects by crossing leaked passport records, travel data and photographs, then matching faces to serving officers.

3 INDEPENDENT PUBLIC SOURCES · 2 ALIASES RESOLVED TO SERVING OFFICERS · 2018 BELLINGCAT AND THE INSIDER

SOURCE · BELLINGCAT AND THE INSIDER INVESTIGATIONS, 2018

06

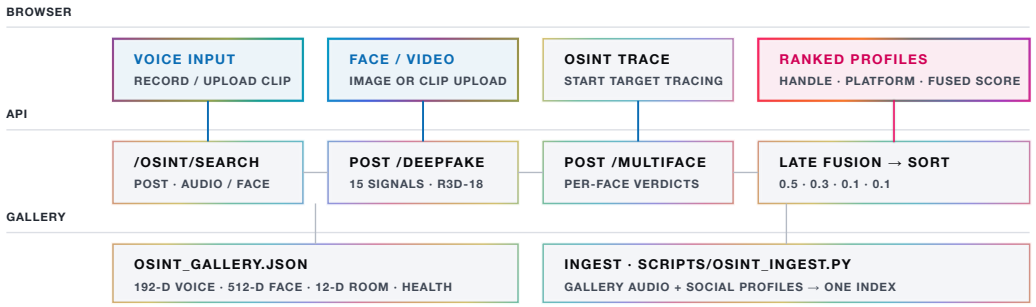
THE LIVE SYSTEM

UPLOAD · EXTRACT · RANK

ACT 6 · THE SERVICE · /DOCS · /OPENAPI.JSON

Upload a clip or a face. Nineteen operations answer.

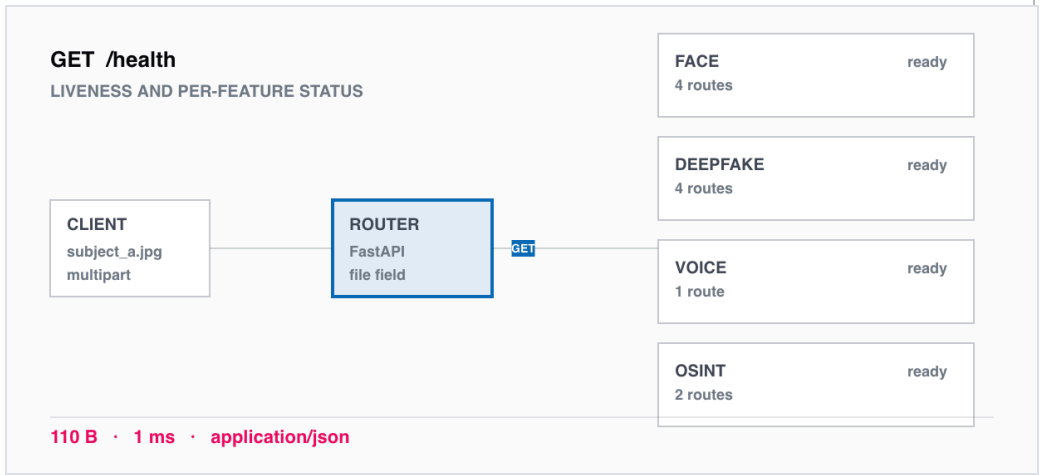
01 THE FLOW



02 THE SURFACE, LIVE

LIVE 127.0.0.1:19123 · LIVE · READY: RESULTS, FACE, DEEPFAKE, VOICE, OSINT · DOWN:

GET	/	774 B	1 ms	GET	/health	120 B	19.29 s
GET	/face/topology	39.6 KB	4 ms	POST	/face/detect	535 B	5 ms
POST	/face/landmarks	31.1 KB	6 ms	POST	/face/mesh	304.7 KB	84 ms
POST	/deepfake	964 B	3.56 s	POST	/deepfake/regions	1.2 KB	19 ms
POST	/deepfake/heatmap	265.3 KB	3.69 s	POST	/deepfake/multiface	736 B	3.50 s
POST	/voice/analyze	2.7 KB	1.87 s	GET	/osint/gallery	1.3 KB	1 ms
POST	/osint/search	500	720 ms	GET	/results	9.7 KB	5 ms
GET	/r/{rid}.json	1.3 KB	1 ms	GET	/r/{rid}.json?embed=1	215.6 KB	2 ms
GET	/r/{rid}	7.0 KB	1 ms	GET	/r/{rid}/assets/{name}	92.1 KB	1 ms
POST	/results	201	0 ms	DELETE	/r/{rid}	0	0 ms



EVERY ROUTE TAKES ONE MULTIPART FIELD. THE PANEL CALLS THE SERVICE AND RE-TIMES THE CHEAP ROUTES IN THE BROWSER. EIGHTEEN OF NINETEEN ANSWER: *POST /OSINT/SEARCH* RETURNS 500, BECAUSE THE GALLERY HOLDS 192-D VOICE EMBEDDINGS AND THE LIVE ENCODER EMITS 256-D. UPLOADS ARE PROCESSED AND DELETED; THE EMBEDDING IS WHAT GETS COMPARED.

POST /DEEPFAKE · /DEEPFAKE/REGIONS · /DEEPFAKE/HEATMAP

Two real people. Both come back FAKE.

01 THE VERDICT, LIVE

LIVE CAPTURED · SUBJECT_A.JPG · 16 FRAMES · CPU · R3D-18 + 15-METHOD SIGNAL ENSEMBLE V4

FAKE 47.33% FAKE PROBABILITY · 45% THRESHOLD · 7 / 9 VOTES · BORDERLINE TIER

16 METHOD SCORES · THE R3D-18 HEAD IS UNTRAINED

FFT ARTIFACT	0.53	SRM NOISE	0.00
CHROMA INCONSISTENCY	0.39	BOUNDARY SEAM	0.00
OPTICAL FLOW	0.00	RPPG	0.00
COLOR JITTER	0.00	LOCAL TEXTURE	0.93
EYE ARTIFACT	0.30	FACIAL SYMMETRY	1.00
TEMPORAL VARIANCE	0.00	GRADIENT DIRECTION	1.00
DOUBLE JPEG	0.72	SKIN REGION CONSISTENCY	0.25
SPECULAR HIGHLIGHT	0.00	R3D 18	0.41

8 REGIONS · POST /DEEPFAKE/REGIONS · WORST FOREHEAD · MEAN 23.2%

FOREHEAD	LEFT EYE	RIGHT EYE	NOSE	LIPS	L. CHEEK	R. CHEEK	CHIN
34% CLEAN	24% CLEAN	18% CLEAN	19% CLEAN	17% CLEAN	28% CLEAN	23% CLEAN	23% CLEAN

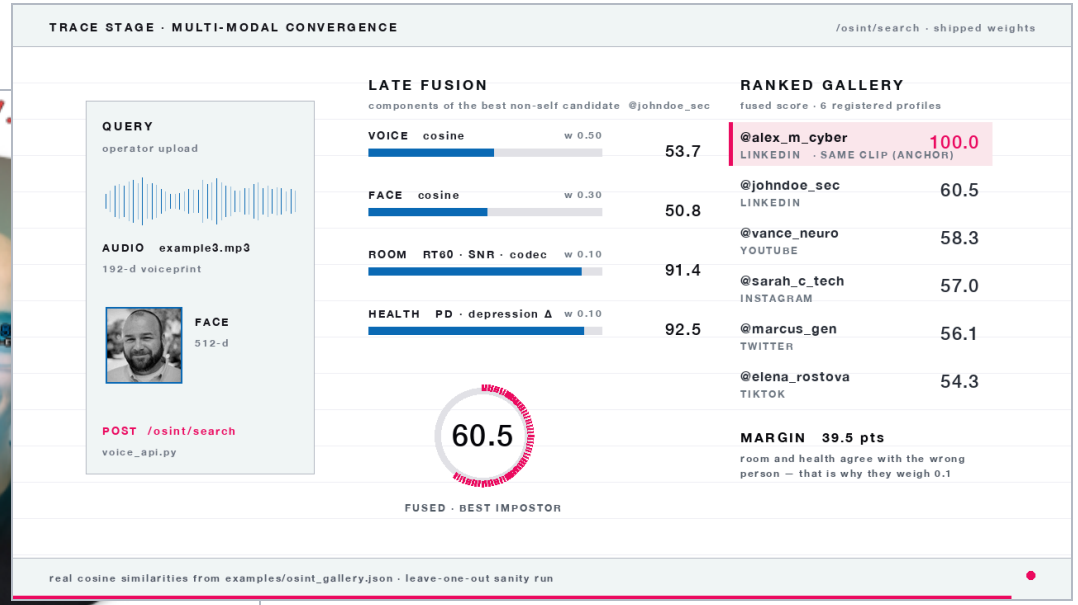


POST /DEEPFAKE/HEATMAP · 720 × 720 PNG

SUBJECT A

02 ONE UPLOAD, SIX CANDIDATES

LIVE TRACE



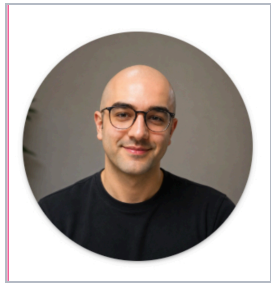
SUBJECT A RETURNS FAKE AT 47.33 %, VOTES SEVEN TO NINE. SUBJECT B RETURNS FAKE AT 45.34 %, VOTES EIGHT TO EIGHT. THE THRESHOLD IS 45 %, BOTH ARE TIER BORDERLINE, AND BOTH ARE GENUINE PHOTOGRAPHS OF REAL PEOPLE.

GET /R/{ID}.JSON?EMBED=1 · 364 KB, THE WHOLE RECORD

One id returns the verdict, the frames, the regions and the pictures.

01 THE STORED RECORD, LIVE

LIVE · STORED DEEPFAKE RECORD · 16 FRAMES · CPU · R3D-18 + 15-METHOD SIGNAL ENSEMBLE V4



SOURCE · HEATMAP · WIPE

FAKE

47.93%	45%	7 / 9	BORDERLINE
FAKE PROBABILITY	THRESHOLD	VOTES	TIER

ONE PHOTOGRAPH, THREE RUNS OF IT	SERVER · /DEEPFAKE	THIS RECORD	GPU + BROWSER
	47.33%	47.93%	56.74%

16 FRAME SCORES · IDENTICAL, BECAUSE A STILL HAS NO MOTION

16 METHOD SCORES

8 REGIONS, INSIDE THE SAME DOCUMENT

FOREHEAD	LEFT EYE	RIGHT EYE	NOSE	LIPS	L. CHEEK	R. CHEEK	CHIN
34% CLEAN	24% CLEAN	18% CLEAN	19% CLEAN	17% CLEAN	28% CLEAN	23% CLEAN	23% CLEAN

WHY IT SAID SO

ONE REQUEST 364 KB FOLLOW-UP CALLS 0

02 THE READOUT STACK

DIARISATION
MFCC+Δ CLUSTERING → ECAPA
HOW MANY VOICES, NOT WHO

SEX · AGE
F0 · CENTROID · LH RATIO
WEIGHTED VOTE · ~95 %
A BAND, NOT A YEAR

HEALTH SCREENING
PARKINSON'S 4 · DEPRESSION 4
0–100 · LOW / MOD / ELEVATED

EMOTION
AROUSAL · VALENCE · DOMIN.
57 EMOTIONS RANKED

LANGUAGE · ACCENT
256 LANGUAGES · META MMS
8 ENGLISH REGIONS · 50–65 %
85–92 % TOP-1, CLEAN >5 S

ENVIRONMENT
RT60 · SNR · CODEC · CLASS
12-D ROOM FINGERPRINT

VOICEPRINT
ECAPA-TDNN · GE2E FALLBACK
256-D L2-NORMALISED
IDENTITY

AUTHENTICITY
15 SIGNALS · R3D-18 W 0.05
RUNS FIRST · GATES THE REST
REAL / FAKE

SIXTEEN METHOD SCORES, SIXTEEN FRAME SCORES, EIGHT REGIONS, THREE STATED REASONS, THE VOTES, THE DEVICE, THE MODEL AND THE HEATMAP, IN ONE RESPONSE. THE SAME PHOTOGRAPH HAS BEEN SCORED THREE TIMES BY THIS SERVICE: 47.33 % ON THE SERVER, 47.93 % IN THIS RECORD, 56.74 % IN THE BROWSER ON THE GPU.

POST /VOICE/ANALYZE · TEST13338.M4A · 4.07 S · AAC MONO 48 KHZ

Four seconds of speech, and the service answers all of it.

● LIVE LIVE · POST /VOICE/ANALYZE · TEST13338.M4A · 5 ACOUSTIC FIELDS

97.4	MALE	ADULT
HZ · MEDIAN F0	SEX FROM VOICE	AGE BAND

PHENOTYPE · REPORTED DIRECTLY BY THE SERVICE

F0 RANGE	120.5 Hz	SPECTRAL CENTROID	1278 Hz
VOICED FRACTION	38.4 %	F0 TREMOR	0.0358

EMOTION · TOP 6 OF 59



8	30	NOSTALGIA
PD POINTS	DEPRESSION POINTS	TOP EMOTION

HEALTH MARKERS · EACH WITH THE RULE IT WAS SCORED BY

VOCAL TREMOR
0.0358
4–8 Hz F0 tremor power ratio (risk > 0.08)
0 POINTS

F0 MONOTONICITY
120.5
duration-scaled F0 range Hz (risk < 40.0 Hz)
0 POINTS

DYSPHONIA
skipped
skipped — lossy compression artifacts
JITTER 0.02567 · SHIMMER 0.17504 · HNR -30 DB

SPEECH RATE
0.384
voiced fraction (risk < 0.4)
8 POINTS

AROUSAL / VALENCE / DOMINANCE · 0.459 · 0.571 · 0.481

FILE **test13338.m4a** 4.07 S **48 kHz mono**

SEX, AGE BAND, MEDIAN F0, 59 EMOTION SCORES AND TWO HEALTH MARKERS, FROM FOUR SECONDS. THE DYSPHONIA MARKER IS THE HONEST ONE: THE SERVICE REFUSES TO SCORE JITTER AND SHIMMER AT ALL, BECAUSE THE FILE IS LOSSY, AND SAYS SO IN THE RESPONSE RATHER THAN RETURNING A NUMBER IT CANNOT STAND BEHIND.

POST /FACE/MESH · 468 VERTICES · 512 PX ATLAS · AND THE RENDER, STORED

The geometry is free. The verdict costs five seconds.

● LIVE LIVE · POST /FACE/MESH · SUBJECT_A.JPG · 468 VERTICES · 886 TRIANGLES · 512 PX ATLAS



VERTICES 468 TRIANGLES 886 YAW -0.2° PITCH 26.0° ROLL 6.5°

● KEPT WITH THE RESULT LIVE · P85-VJTCXB… · AVATAR_1.JPG


SOURCE.JPG
15.9 KB


HEATMAP.PNG
137.6 KB


MESH.PNG
709.6 KB

TWO OF THE FIFTY KEEP ALL THREE · THE RENDER IS STORED, NOT RECOMPUTED

MESH, ATLAS, 68 LANDMARKS, IRIS POINTS, POSE AND ELEVEN RATIOS IN 84 MS, AGAINST 3.6 S FOR A VERDICT ON CPU. THE RENDER IS STORED, NOT RECOMPUTED.

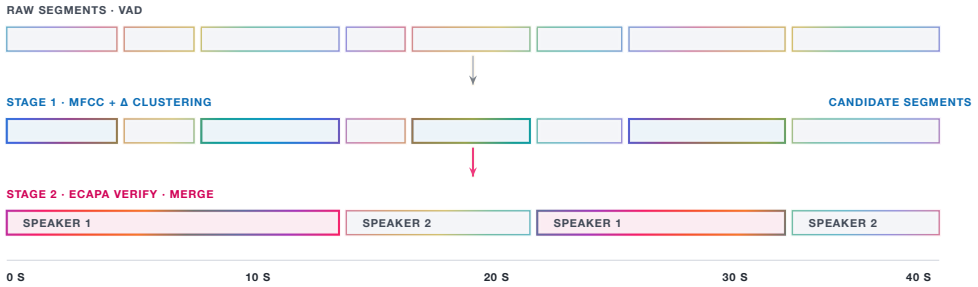
READOUT · REAL VALUES

The margin is the finding, and two stages make it findable.

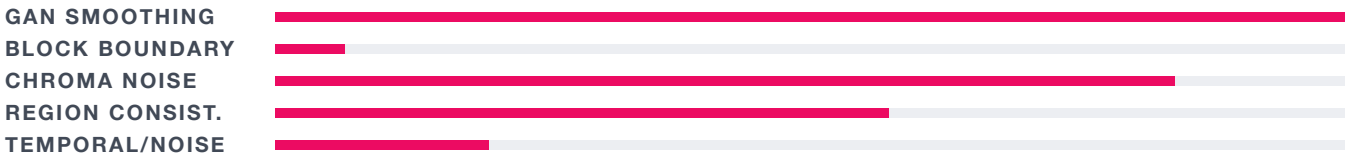
01 THE MARGIN

PROFILE		VOICE	FACE	ROOM	HEALTH	FUSED
@alex_m_cyber	LINKEDIN · SAME CLIP	100.0	100.0	100.0	100.0	100.0
@johndoe_sec	LINKEDIN	53.7	50.8	91.4	92.5	60.5
@vance_neuro	YOUTUBE	50.8	55.7	79.2	82.5	58.3
@sarah_c_tech	INSTAGRAM	49.6	52.7	81.6	82.5	57.1
@marcus_gen	X	49.6	49.7	81.6	82.5	56.1
@elena_rostova	TIKTOK	55.8	47.2	80.2	42.5	54.3

02 DIARISATION



● FROM THE API LIVE · GET /R/WAESDPLIDC….JSON · RESULT.RESTORE



3 / 2
FACES FOUND / USED

57%
ON THE SCREEN

face-crop
CROP MODE

0.4807
GPU SCORE

still
MEDIA KIND

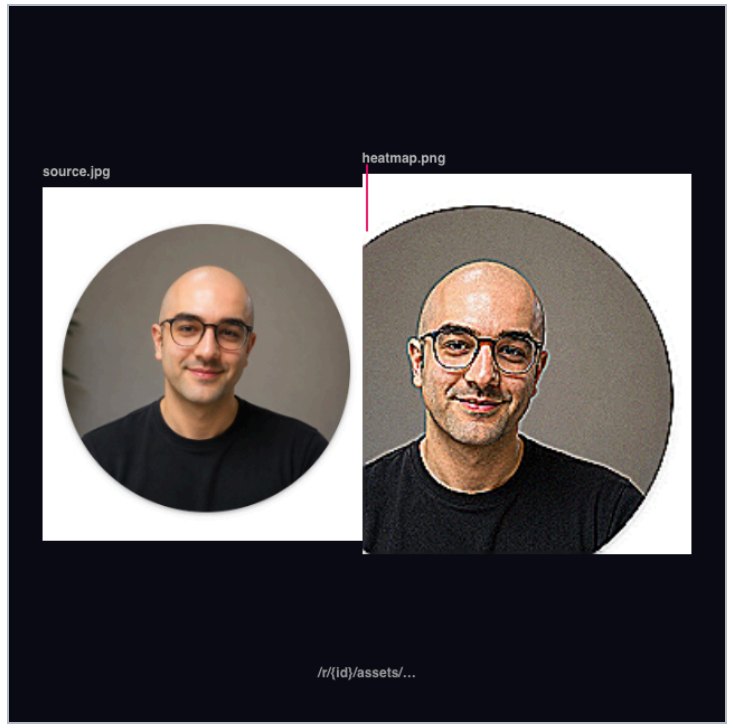
gpu+browser
PIPELINE

THE FIVE BARS THE OPERATOR READS ARE RESULT.RESTORE.MS, TO FOUR DECIMAL PLACES. THE SELECTION, THE CROP MODE AND THE PROBABILITY ARE IN THE SAME RECORD. NOTHING ON THAT SCREEN IS DRAWN FROM ANYWHERE ELSE CLUSTERING PASS ALONE.

WORKED EXAMPLE · FROM THE RUNNING SERVICE

A result keeps the image it was run on.

● LIVE CAPTURED · GET /RESULTS · 50 OF 50 CARRY IMAGES



STORED RESULTS, WITH THE IMAGE EACH ONE WAS RUN ON

A RESULT KEEPS ITS EVIDENCE

50
RESULTS

50
WITH IMAGES

3
KINDS

1
DELETED EVER

thirteen of the fifty carry the source image and the heatmap that was drawn on it. the analysis is reproducible because the input never left.

ALSO, IN THE WORLD
· 2020 TO 2024

The same product, already shipped.

Clearview AI built a face search engine from tens of billions of scraped images and sold access to police forces: upload a face, get a ranked list of people.

30 BN+ IMAGES CLAIMED · €30.5 M DUTCH FINE, 2024 · €20 M FRANCE AND ITALY EACH, 2022

SOURCE · CNIL, GARANTE, HELLENIC DPA AND DUTCH DPA DECISIONS, 2022 TO 2024

07

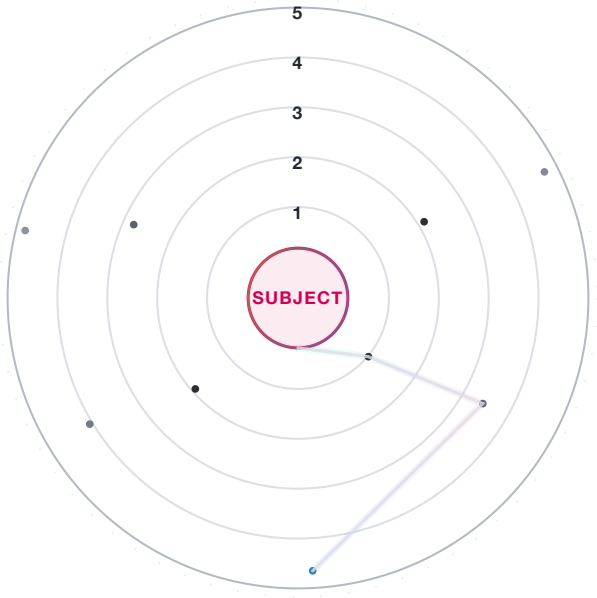
REACH

IDENTIFIED IS ONE STEP · REACHABLE IS THE NEXT

ACT 7 · GRAPH DISTANCE

Six hops from one subject, and the graph cuts both ways.

01 SIX DEGREES



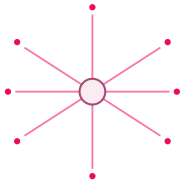
ILLUSTRATIVE BRANCHING FACTOR 150 · 150 → 2.2×10^{4} → 3.4×10^{6} → 5.1×10^{8} → SATURATION

02 BOTH WAYS

AUTHENTIC · ORGANIC



SYNTHETIC · SEEDED



TEMPLATE REUSE	94 %
POST BURST VELOCITY	41 ×
ACCOUNT AGE < 14 D	88 %
FACE REUSE ACROSS HANDLES	detected

ILLUSTRATIVE DETECTION SIGNATURE · NO LIVE ACCOUNTS. THE NETWORK THAT FINDS A PERSON IS THE SAME SURFACE AN INFLUENCE OPERATION DELIVERS ON — THIS IS WHAT DETECTION HAS TO CATCH.

SIX DEGREES OF SEPARATION

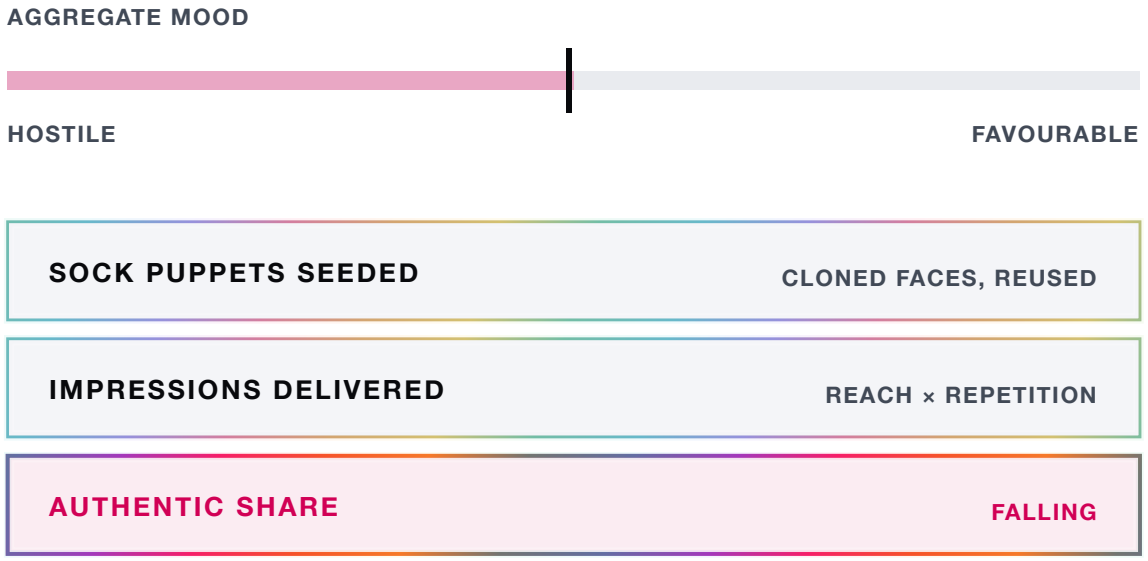
EVERYONE IS ≤ 6 CONNECTIONS AWAY

● DEGREE 6 / 6 MOST OF A POPULATION

ILLUSTRATIVE · FICTIONAL TARGET

Then you stop tracing people and start steering them.

Identification is the entry step, not the objective. The objective is the aggregate — and the aggregate has a needle.



FICTIONAL TARGET · NO LIVE ACCOUNTS · SHOWN SO DETECTION CAN BE BUILT AGAINST IT

OPERATION 00 · INFILTRATE

Seeding synthetic accounts

NETWORK PENETRATION 55%

FICTIONAL TARGET · ILLUSTRATIVE ONLY

NOW STEER WHAT THEY FEEL

SYNTHETIC CROWDS · MANUFACTURED CONSENSUS · SENTIMENT AT SCALE

AGGREGATE SOCIETY MOOD

TILTING

HOSTILE — FAVOURABLE

SOCK PUPPETS	816
IMPRESSIONS	6.3 M
AUTHENTIC SHARE	86.8%

CAMPAIGN OBJECTIVES — RUNNING

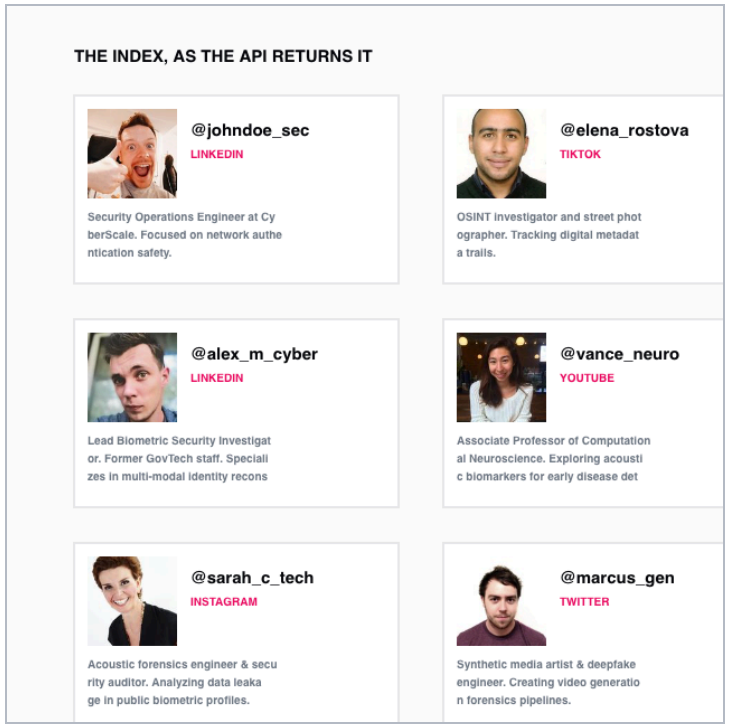
INDIVIDUAL · PUBLIC STANDING	▼ 86%
HOUSING MARKET · PURCHASE INTENT	▲ 83%
FEATURE RELEASE · AUDIENCE SCORE	▲ 78%
INVESTIGATION · STORY REACH	▼ 72%
OUT-GROUP · HOSTILITY INDEX	▲ 66%
CANDIDATE · FAVOURABILITY	▲ 61%

ILLUSTRATIVE THREAT MODEL · NO LIVE ACCOUNTS · WHAT DETECTION MUST CATCH

WORKED EXAMPLE · FROM THE RUNNING SERVICE

Six handles, five platforms, one index.

● LIVE CAPTURED · GET /OSINT/GALLERY · PLATFORMS



SIX HANDLES, FIVE PLATFORMS, ONE INDEX

THE REACH A FARM IS BUILT TO FEED

6
HANDLES

5
PLATFORMS

0
SEARCHES RUN

6
FACES IN THE DEMO FARM

the gallery is small because it is an example. the mechanism is not: one face, many handles, and a score that says which of them is the same person.

ALSO, IN THE WORLD
· 2022 TO 2023

A rack of phones is a crowd.

The cheapest way to look like a thousand people is to be a thousand handsets.

≈500 PHONES IN ONE RAID, THAILAND 2022 · 350,000 SIM CARDS, SAME SEIZURE · 7,700 ACCOUNTS IN META'S LARGEST INFLUENCE TAKEDOWN

SOURCE · THAI POLICE REPORTING, 2022; META ADVERSARIAL THREAT REPORT, 2023

08

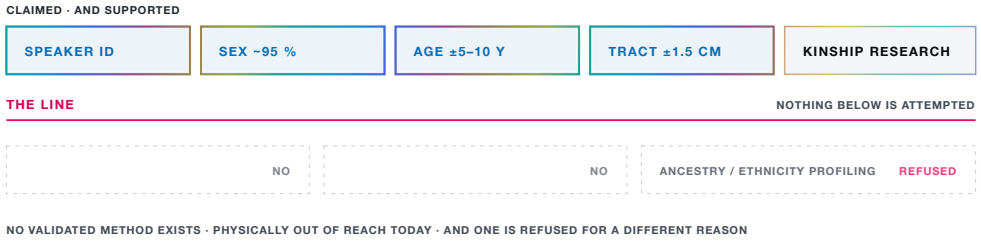
THE LINE

WHAT IS REAL · WHAT IS REFUSED

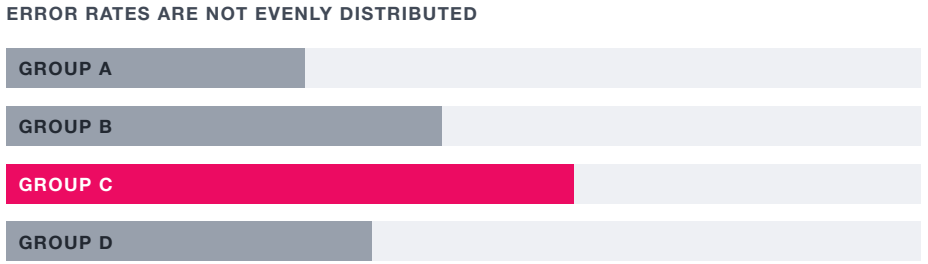
ACT 8 · THE LINE

What is above the line, and what is refused on bias.

01 ABOVE THE LINE



02 REFUSED



GDPR ARTICLE 9	BIOMETRIC SPECIAL-CATEGORY DATA
ILLINOIS BIPA	WRITTEN CONSENT REQUIRED
CCPA	DISCLOSURE AND DELETION RIGHTS

NO DECISION AFFECTING AN INDIVIDUAL SHOULD REST ON AUTOMATED VOICE OR FACE MATCHING ALONE. OBTAIN LEGAL ADVICE BEFORE OPERATIONAL USE.

SELF-REJECTION

The second paper, and what a reviewer would reject it for.

01 PAPER II

IN-DOMAIN

0.91

AUC · SAME CORPUS

CROSS-DATASET

16.42 %

EER · UNSEEN CORPUS

WHAT TRANSFERS

RANKING WITHIN ONE CORPUS · RELATIVE ORDERING OF CANDIDATES

WHAT DOES NOT

ABSOLUTE THRESHOLDS · CALIBRATION · THE HEADLINE NUMBER

TRAIN ON ONE, TEST ON ANOTHER, PUBLISH THE WORSE FIGURE

02 THE REJECT LIST

KINSHIP HEAD
MISSING — THE CLAIM THAT EARNS IT

BASELINES
NO X-VECTOR · NO ECAPA · NO 2D CONTROL

MINDCF
EER ALONE IS HALF THE STORY

PHENOTYPE HEADS
SEX / AGE / VTL — PLANNED, NOT TRAINED

GA ABLATION
VS UNIFORM AND VS LEARNED WEIGHTS

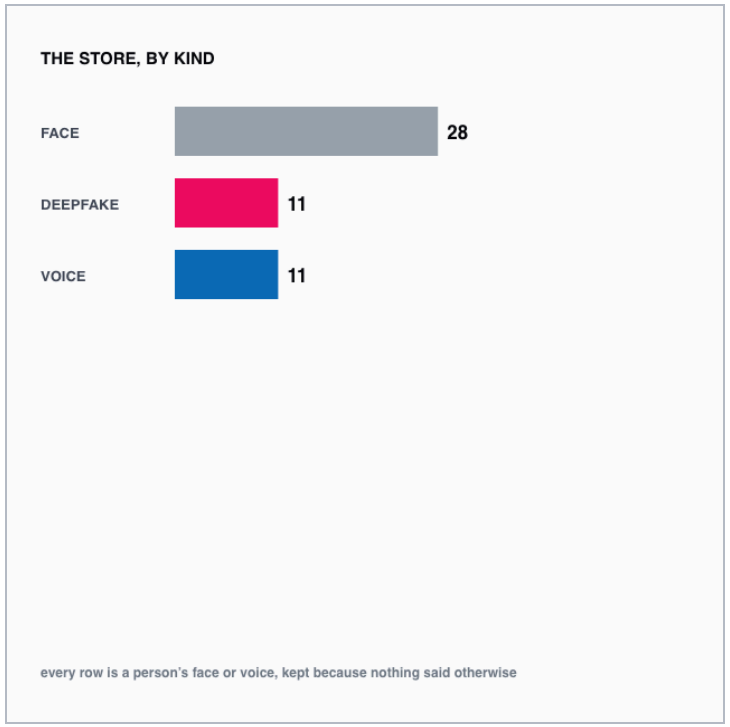
PRIVACY FRAMEWORK
CONSENT · BIAS AUDIT · SAFEGUARDS

VOXCELEB1-O VERIFICATION LIST · CROSS-DATASET EVAL · PER-GROUP ERROR RATES
LISTED BECAUSE THEY ARE TRUE — A SYSTEM THAT PUBLISHES ITS OWN REJECTION CRITERIA IS EASIER TO TRUST

WORKED EXAMPLE · FROM THE RUNNING SERVICE

Fifty results, no expiry, no auth.

● LIVE CAPTURED · GET /RESULTS · THE STORE, UNPRUNED



FIFTY RESULTS, NONE OF THEM EXPIRED

WHAT IS STILL ON DISK

50
RESULTS KEPT

50
WITH IMAGES

none
RETENTION POLICY

0
DELETED BY THIS DECK

there is no expiry, no auth and no audit on the store. DELETE /r/{id} exists and works. nothing in this deck calls it, and nothing in the service requires anyone to.

ALSO, IN THE WORLD
· 2021 TO 2025

The line is now written down.

The EU AI Act entered into force in August 2024 and its prohibitions applied from February 2025, among them building face recognition databases by untargeted scraping.

$650 M BIPA SETTLEMENT, 2021 · 1 AUG 2024 EU AI ACT IN FORCE · 2 FEB 2025 PROHIBITIONS APPLY

SOURCE · REGULATION (EU) 2024/1689; IN RE FACEBOOK BIOMETRIC INFORMATION PRIVACY LITIGATION

GENETRACE · BIOMETRIC INTELLIGENCE

Trace anyone

15-METHOD DEEPFAKE DETECTION · OSINT GRAPH · GENETIC-PHENOTYPE PROFILING

▱ ARXIV PAPER

</> SOURCE CODE

ENTER SYSTEM →

"GENETIC" = HERITABLE VOICE/FACE PHENOTYPE (SEX, AGE, VOCAL-TRACT SIZE) + A GENETIC-ALGORITHM MATCHER. RESEARCH PROTOTYPE.

ROMHACK

Documented for research.
Not for hunting people.

The capability is real either way. Publishing what it can do — and what it refuses — is the only part of that a builder controls.

ARXIV 2605.17573 | ARXIV 2606.29506 | DEMO VOICE_DASHBOARD.HTML